**From:** **Leonardo Dayal** ldayal@asu.edu
**Subject:** Re: Additional Measurements
**Date:** November 9, 2026 at 11:34
**To:** Samantha Taylor Samantha.Kneale@asu.edu
**Cc:** Marcus Herrmann mherrma1@asu.edu, Justin Flory Justin.Flory@asu.edu

Hey Samantha!
Thanks so much for the information on the general geometry!

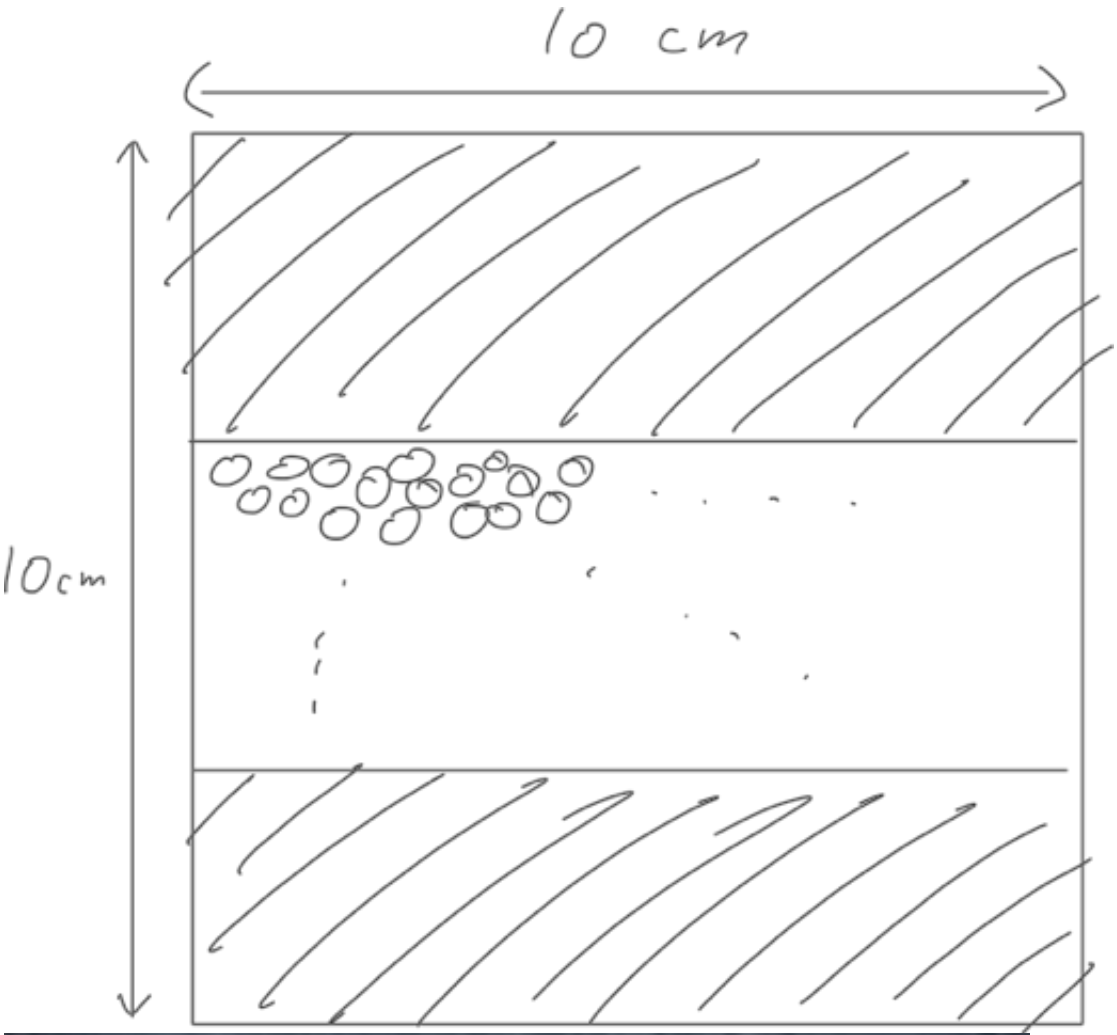
One quick clarification: you described the thermo-sealed line as solid. Can you confirm it is airtight, with no airflow through it?

Marcus and I had an idea of a workaround for the blob issues you mentioned in the email. If the flow is blocked everywhere except the bead-filled areas (see picture 1) we can see how the beads impact flow independent of packet geometry. The slashed areas could be some kind of 3d printed cover or, if the seals are airtight, thermo-sealing the entire top and bottom. Running it once empty and once filled then accounts for the set-up itself, so we are only capturing the effect of the beads.

In the meantime, I am going to do a proof of concept on some calibration with the data you sent on Nov 5. Since those packets are not super full (picture 2), the results might not be very accurate. If we do settle on a design where they are not fully filled, I might need to add some more complexity to the solver by accounting for the unevenness.

Let me know what you think and thanks again!

Kind Regards,
Leo








On Nov 6, 2026, at 17:34, Samantha Taylor <Samantha.Kneale@asu.edu> wrote:

Hi Leonardo,

1. They are more similar to rods, bulging in both directions.
2. That measurement (t) is very dependent and the amount of sorbent fill. I can get you a measurement (t) on Monday for the one Ramesh ran (0.3 g fill). Justin has suggested we also run a test with the stick style packet before filling (with the bars in it but no sorbent), as well as a fully packed run (as much sorbent as we can get in each stick). I will get you t measurements for that as well.
3. Distance between them (d) is a solid thermo-sealed line of 2mm or ~ 0.079 inch

The bars/sticks exist because the sorbent beads settle, so there isn't really any way to get a full packet in a tidy, consistent thickness without some kind of quilting/structure. I can do the fully packed version with the sticks as suggested above, but a single pocket packets going to be a blob in the wind tunnel.

The beads do swell when wet. We should be able to do a run at a higher humidity, but it'll be dependent on the wind tunnels capabilities which I am not sure of at the moment. I will ask Ramesh and Lindsey Hamblin if they have a max we can hold at.

I'll work on this Monday but won't be able to run any wind tunnel pressure drop tests until Ramesh returns from AICHE the following week.

Thanks and have a lovely weekend!

**Samantha Kneale Taylor**
Laboratory & Research Manager
Center for Negative Carbon Emissions

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**From:** Leonardo Dayal <ldayal@asu.edu>
**Sent:** Friday, November 6, 2026 4:45 PM
**To:** Samantha Taylor <Samantha.Kneale@asu.edu>
**Cc:** Marcus Herrmann <mherrma1@asu.edu>; Justin Flory <Justin.Flory@asu.edu>
**Subject:** Fwd: Additional Measurements

Hey Samantha!

Me and Dr. Herrmann were talking today and realized that we still need some more details on the exact measurements!

1. What is the exact shape of the bead packets themselves is it like this semicircular shape or more circular like rods:

<image001.png>

2. What's the thickness (t) when they are filled up
3. How far are they apart from each other (d)?

We also talked about additional tests we need to figure out the coefficients in our model. Specifically, we would like a test where the beads are just between two sheets without making it into these "bars":

<image002.png>

This will give us a consistent flow to calibrate everything and then compare our model to the bar baseline that you measure.

Finally, I could be mistaken on this but did you mention that the beads swell up with more moisture content? If so we might need more data regarding specific pressure drops with moisture as well

Thank you so much and let me know if you have any questions!

Kind Regards,
Leo

Ps. I realized that I sent this email to the wrong Samantha! Sorry about that!